



React 19 完整学习指南

🎯 面试星级：★★★★★ | 建议用时：3 天

React 19 系统学习指南，融合核心原理、高级特性、工程实践与面试题，从入门到精通



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第1章：起航React之旅

1-1 全景透视：React 课程精华与学习决胜路径

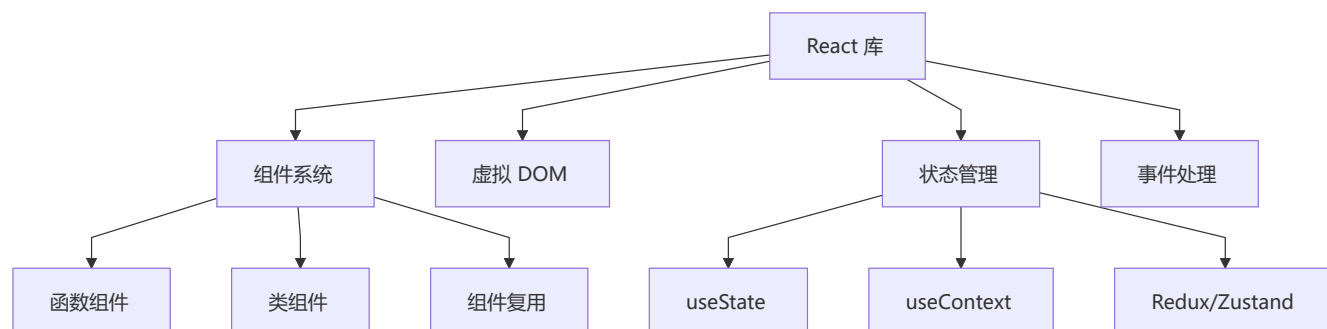


React 的核心定义

React 是由 Facebook (Meta) 开发的 JavaScript 库，用于构建用户界面。它通过**组件化思想**和**声明式编程**，帮助开发者高效构建交互式、动态的 Web 应用。

```
// React 的三大特性：  
// 1. 声明式：描述你想要什么，而不是如何实现  
// 2. 组件化：封装独立可复用的 UI 单元  
// 3. 虚拟 DOM：高效批量更新真实 DOM
```

React 的核心角色



React 五大设计理念

React 的设计哲学可以概括为 "UI = f(state)" — 视图是状态函数的输出。

① 声明式 (Declarative)

```
// ❌ 命令式 (jQuery 思维)
const div = document.createElement('div');
div.className = 'card';
div.textContent = 'Hello';
parent.appendChild(div);

// ✅ 声明式 (React 思维)
function Card({ text }) {
  return <div className="card">{text}</div>;
}
```

② 组件化 (Component-Based)

```
// 组件 = 独立单元
function UserCard({ user }) {
  return (
    <Card>
      <Avatar src={user.avatar} />
      <Name>{user.name}</Name>
      <Stats posts={user.postCount} />
    </Card>
  );
}
```

③ 虚拟 DOM (Virtual DOM)

状态变化 → 新虚拟 DOM → Diff(旧虚拟DOM, 新虚拟DOM) → Patch(真实DOM)

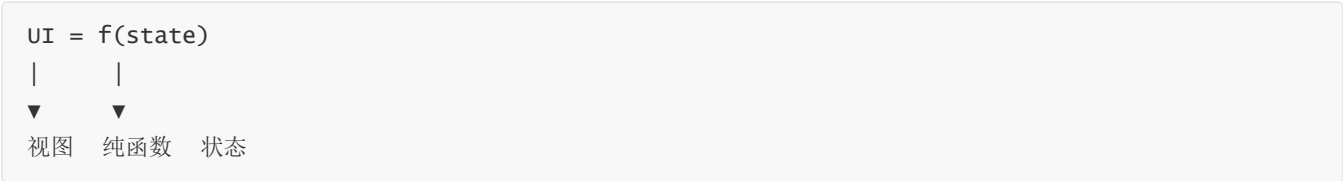
④ 函数式编程 (Functional)

```
// ✅ 不可变数据
function GoodList({ items }) {
  return <ul>{[...items, 'new item'].map(/* ... */)}</ul>;
}
```

⑤ 一次学习，随处编写

- React DOM → Web 应用
- React Native → iOS / Android 原生应用
- React Three → 3D 场景 (Three.js 封装)
- React Ink → 命令行终端 UI

💡 一个公式理解 React



📊 React vs 其他框架

特性	React	Vue	Angular
学习曲线	🟡 中等	🟢 平缓	🔴 陡峭
灵活性	✅ 极高	⚠️ 中等	❌ 受限
生态系统	✅ 最庞大	⚠️ 中等	✅ 完整
性能	✅ 优秀	✅ 优秀	✅ 优秀
企业应用	✅ 完美	⚠️ 可行	✅ 完美

📖 课程学习路径

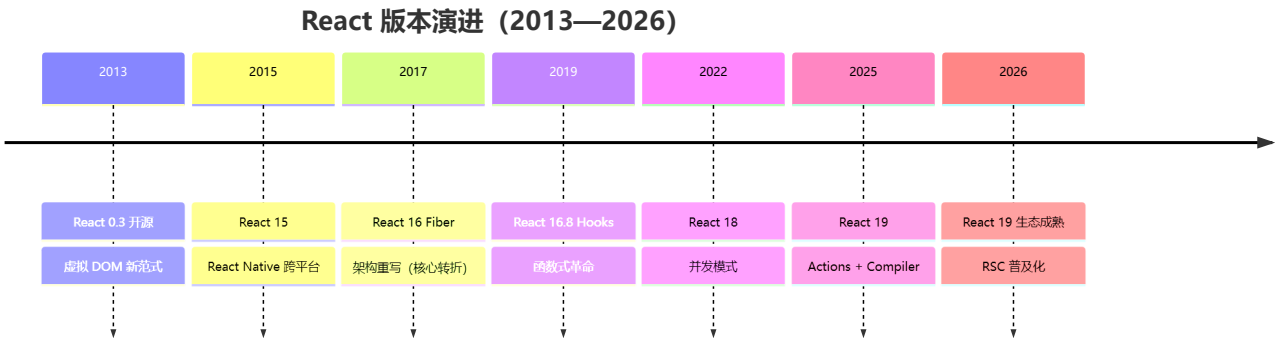




第2章：React 19初体验与开发环境

2-1 前沿洞察：React 领跑前端开发的核心优势

React 版本迭代史（2013—2026）



关键版本逐代解析

版本	年份	核心变化	对开发者的影响
React 0.3	2013	虚拟 DOM, JSX 首次开源	开创性范式：声明式 UI
React 15	2016	DOM 重构 + React Native	跨平台能力
React 16	2017	Fiber 架构重写	可中断渲染, 优先级调度
React 16.8	2019	Hooks 发布	函数组件拥有状态
React 18	2022	并发模式、自动批处理	Suspense 完善
React 19	2025	Actions、use()、React Compiler	表单革新、自动记忆化

React 18 → 19 核心变化

特性	React 18	React 19
并发模式	可选启用	默认启用
Actions	✗	✓ 统一表单处理
use()	✗	✓ 异步数据获取
useOptimistic	✗	✓ 乐观更新
Server Components	实验性	✓ 稳定
ref 传参	forwardRef	直接传 ref
React Compiler	实验性	✓ 自动 memo

2-2 一步到位：React 开发环境搭建实战全攻略

环境要求

```
Node.js 18+  
npm / yarn / pnpm  
VS Code + ESLint + Prettier 插件
```

2-3 Vite + React 19 闪电战：3秒启动

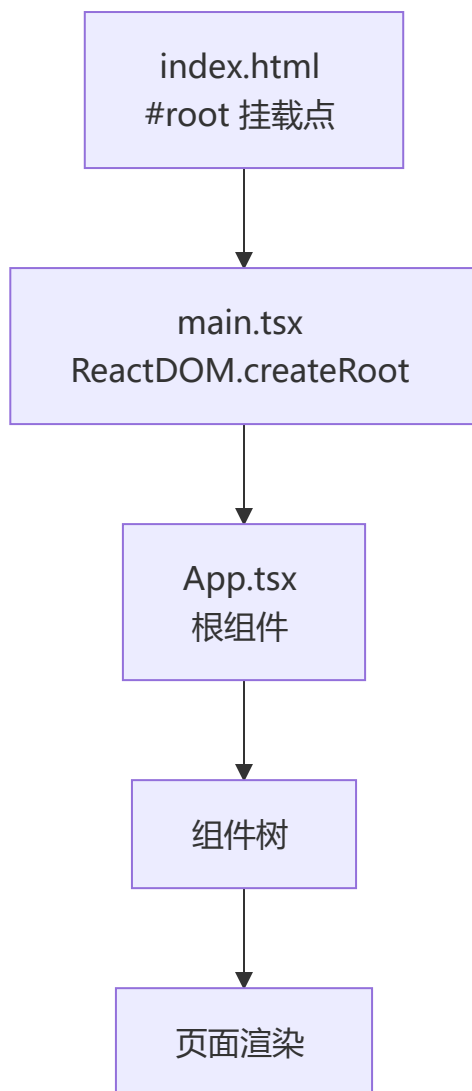
创建项目

```
npm create vite@latest my-react-app -- --template react-ts  
cd my-react-app  
npm install  
npm run dev
```

目录结构

```
my-react-app/  
├─ index.html      # 入口 HTML  
├─ src/  
│  ├─ main.tsx    # 应用入口  
│  ├─ App.tsx     # 根组件  
│  ├─ App.css  
│  ├─ index.css  
│  ├─ assets/  
│  └─ vite-env.d.ts  
├─ package.json  
├─ tsconfig.json  
├─ vite.config.ts  
└─ public/
```

2-4 Vite+React 项目核心结构剖析



2-5 项目总控：package.json

```
{
  "name": "my-react-app",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "tsc && vite build",
    "lint": "eslint .",
    "preview": "vite preview"
  },
  "dependencies": {
    "react": "^19.0.0",
    "react-dom": "^19.0.0"
  },
  "devDependencies": {
    "@types/react": "^19.0.0",
```

```
"@types/react-dom": "^19.0.0",
"@vitejs/plugin-react": "^4.0.0",
"typescript": "^5.0.0",
"vite": "^6.0.0"
}
}
```

2-6 工程监理：package-lock.json

`package-lock.json` 锁定依赖版本，确保团队和 CI/CD 环境安装一致的依赖版本，避免"我机器上能跑"的问题。

2-7 施工蓝图：vite.config.js

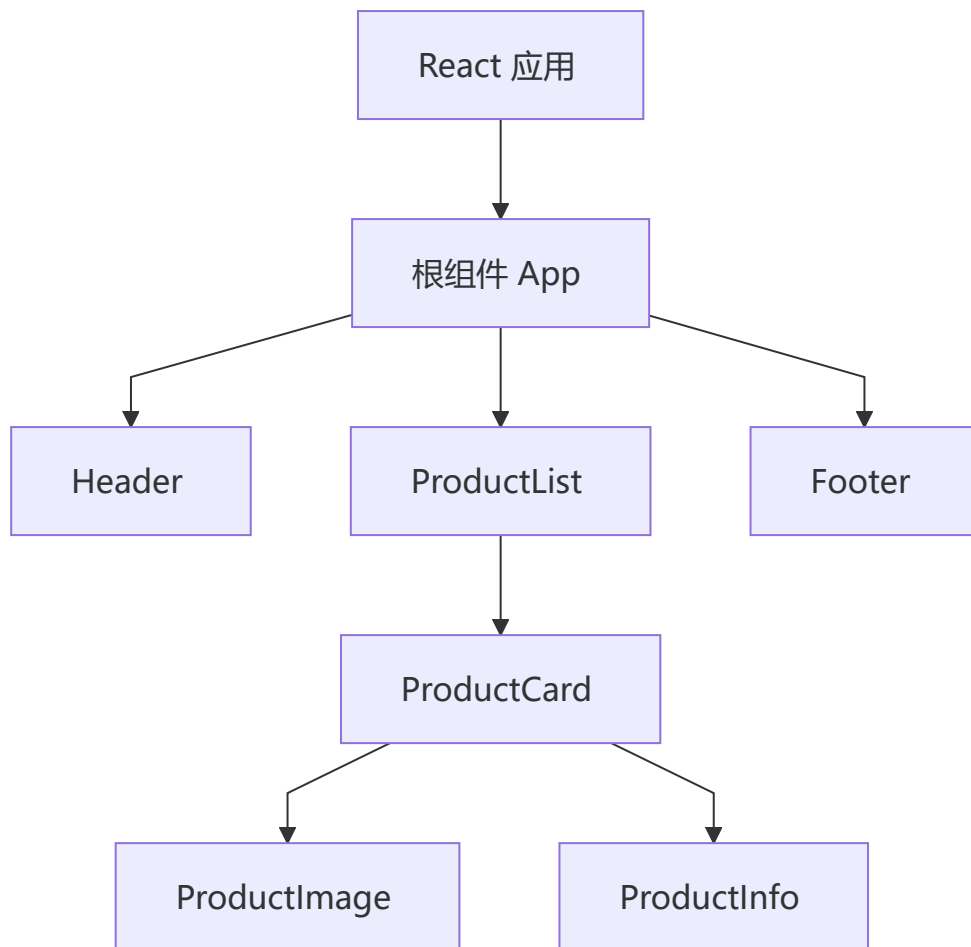
```
import { defineConfig } from 'vite';
import react from '@vitejs/plugin-react';
import path from 'path';

export default defineConfig({
  plugins: [react()],
  resolve: {
    alias: {
      '@': path.resolve(__dirname, './src'),
    },
  },
  server: {
    port: 3000,
    open: true,
  },
});
```



第3章：核心机制与组件化艺术

3-1 组件化架构与 Props 高效通信



3-2 React 根组件：从零构建应用

```
// main.tsx
import { StrictMode } from 'react';
import { createRoot } from 'react-dom/client';
import App from './App.tsx';

createRoot(document.getElementById('root')!).render(
  <StrictMode>
    <App />
  </StrictMode>
);
```

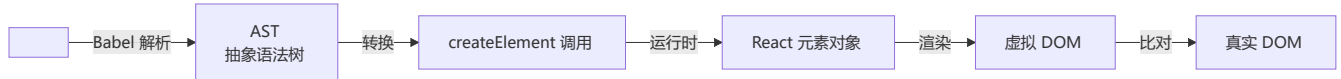
3-3 JSX 语法革命：从 HTML 到 React

JSX 是 JavaScript XML，是 `React.createElement` 的语法糖。


```
// 原始 JSX
const element = <h1 className="greeting">Hello, {name}!</h1>;

// Babel 编译后
const element = React.createElement(
  "h1",
  { className: "greeting" },
  "Hello, ", name, "!"
);
```

JSX 转换流程图



JSX 规则

```
// ✓ 使用 Fragment
return (
  <>
    <p>Hello</p>
    <p>World</p>
  </>
);

// ✓ 属性驼峰命名
<div className="card" data-testid="card" />

// ✓ 表达式插值
<p>Count: {count * 2}</p>

// ✓ 条件渲染
{showTitle ? <h1>Title</h1> : null}
{showTitle && <h1>Title</h1>}
```

3-4 JSX + 组件 + 单根法则

组件必须返回**单个根元素**（Fragment 不算额外 DOM 节点）。

```
// ✗ 错误：多个根元素
function Bad() {
  return (
    <h1>Title</h1>
    <p>Content</p>
  );
}

// ✓ 正确：Fragment 包裹
function Good() {
  return (
    <>
```

```
    </>
    <h1>Title</h1>
    <p>Content</p>
  </>
);
}
```

3-5 React 视觉工程入门：CSS 基础与原子化设计

CSS 引入方式

```
// 方式1: 全局 CSS
import './App.css';

// 方式2: CSS Modules
import styles from './Card.module.css';

// 方式3: 内联样式
<div style={{ color: 'red', fontSize: '16px' }} />

// 方式4: Styled-components
import styled from 'styled-components';
```

3-6 CSS 模块化实战

```
/* Card.module.css */
.card {
  border: 1px solid #e0e0e0;
  border-radius: 8px;
  padding: 16px;
}

.title {
  font-size: 18px;
  font-weight: 600;
}
```

```
import styles from './Card.module.css';

function Card({ title, children }) {
  return (
    <div className={styles.card}>
      <h2 className={styles.title}>{title}</h2>
      {children}
    </div>
  );
}
```

CSS 方案对比

方案	作用域	动态样式	运行时	Bundle
原生 CSS	全局	✗	无	0
CSS Modules	组件级	✗	无	0
Styled-components	组件级	✔ 主题	✔ 有	~15KB
Tailwind CSS	全局	✔ 条件	无	可 tree-shake

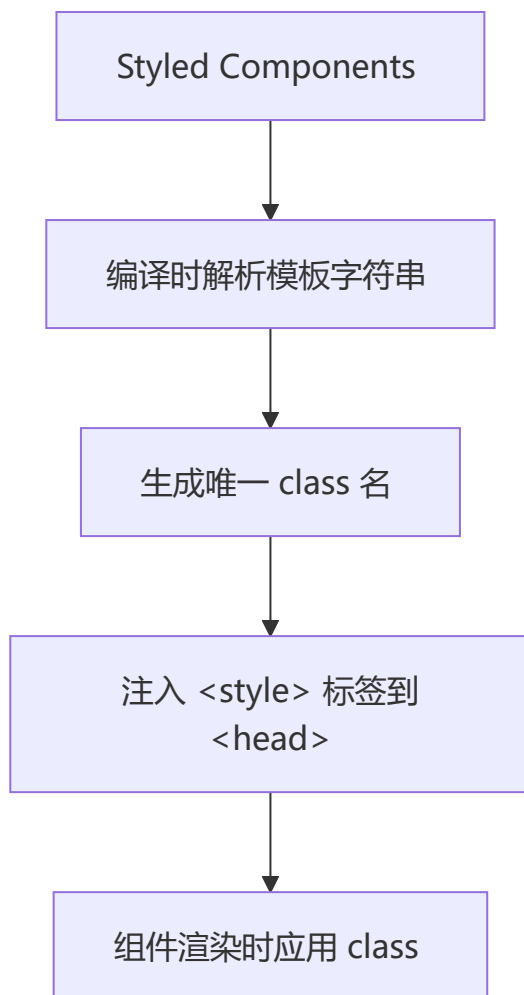
3-7 Styled-components 样式化组件

```
import styled, { css } from 'styled-components';

const Button = styled.button<{ $primary?: boolean }>`
  padding: 12px 24px;
  border-radius: 6px;
  font-size: 16px;
  border: none;
  cursor: pointer;

  ${({ $primary }) => $primary && css`
    background: #007bff;
    color: white;
    &:hover {
      background: #0056b3;
    }
  `}
`;
```

3-8 CSS-in-JS 原理深度拆解



3-9 Props 传声筒：父子孙跨层级数据通道

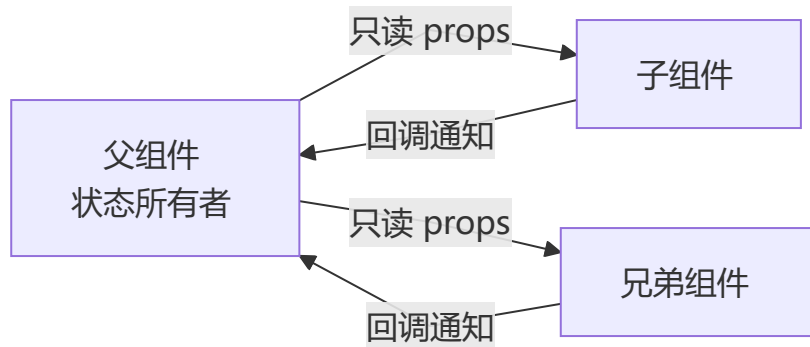
```
interface Product {
  id: number;
  name: string;
  price: number;
}

interface ProductCardProps {
  product: Product;
  onAddToCart: (id: number) => void;
}

function ProductCard({ product, onAddToCart }: ProductCardProps) {
  return (
    <div className="product-card">
      <h3>{product.name}</h3>
      <p>¥{product.price}</p>
      <button onClick={() => onAddToCart(product.id)}>加入购物车</button>
    </div>
  );
}
```

```
}
```

3-10 Props 单向数据流：不可变性的防崩溃设计哲学



⚠ Props 不可变性原则

```
// ❌ 错误：直接修改 props
function Bad({ user }) {
  user.name = 'new name'; // 直接修改 props!
  return <div>{user.name}</div>;
}

// ✅ 正确：props 只读
function Good({ user }) {
  return <div>{user.name}</div>;
}
```

3-11 如何防止 props 被非法修改？用 ESLint 保驾护航

```
{
  "rules": {
    "react/no-direct-mutation-state": "error",
    "react/forbid-component-props": ["error", { "forbid": ["style"] }]
  }
}
```

3-12 Props 黑科技

```
// 解构 + 默认值
function Card({ title = '默认标题', children }) {
  return (
    <div>
      <h2>{title}</h2>
      {children}
    </div>
  );
}
```

```
// Spread 批量传递
const productProps = { name: '手机', price: 2999, stock: 10 };
<ProductCard {...productProps} />

// 回调呼叫（子→父通信）
function Parent() {
  const handleChildEvent = (data: string) => console.log(data);
  return <Child onAction={handleChildEvent} />;
}
```

第4章：高效渲染与组件性能突围

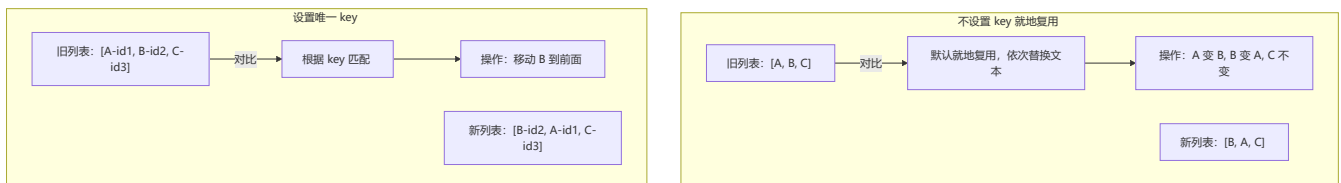
4-1 React 实战之路：从列表渲染到高阶组件



4-2 列表渲染工业级实践

```
function ProductList({ products }: { products: Product[] }) {
  return (
    <div className="grid">
      {products.map(product => (
        <ProductCard key={product.id} product={product} />
      ))}
    </div>
  );
}
```

4-3 Key 属性玄机：Diff 算法核心优化



⚠️ **不要使用 index 作为 key!** 列表发生变化（增删排序）时，index 会变化，导致组件状态错乱。

4-4 短路运算 (&&)：极简逻辑控制

```
function UserProfile({ user }) {
  return (
    <div>
      <h2>{user.name}</h2>
      {user.isVIP && <span className="badge">VIP 用户</span>}
      {/* ⚠️ 注意: user.isVIP 为 0 或 '' 时会渲染 0 或空字符串 */}
      {user.bio && <p>{user.bio}</p>}
    </div>
  );
}
```

4-5 三元运算符：JSX 中的优雅分支决策

```
function StatusMessage({ status }) {
  return (
    <div>
      {status === 'loading' ? <Spinner /> :
      status === 'error' ? <Error /> :
      status === 'success' ? <Success /> :
      null}
    </div>
  );
}
```

4-6 if 的多重返回：组件逻辑分叉与拆分

```
function Dashboard({ user, data, loading, error }) {
  if (loading) return <LoadingSkeleton />;
  if (error) return <ErrorFallback message={error} />;
  if (!user) return <LoginPrompt />;
  if (!data.length) return <EmptyState />;

  return <DataGrid data={data} />;
}
```

4-7 React 组件组合

```
function Layout({ header, sidebar, main }) {
  return (
    <div className="layout">
      <header>{header}</header>
      <aside>{sidebar}</aside>
      <main>{main}</main>
    </div>
  );
}
```

```
// 使用
<Layout
  header={<Header />}
  sidebar={<Sidebar />}
  main={<MainContent />}
/>
```

4-8 props.children：万能插口

```
interface CardProps {
  title: string;
  children: React.ReactNode;
  footer?: React.ReactNode;
}

function Card({ title, children, footer }: CardProps) {
  return (
    <div className="card">
      <div className="card-header">{title}</div>
      <div className="card-body">{children}</div>
      {footer && <div className="card-footer">{footer}</div>}
    </div>
  );
}
```

4-9 具名插槽：组件组合的精准对接

```
function ProductCard({
  image,
  info,
  actions,
}: {
  image: React.ReactNode;
  info: React.ReactNode;
  actions: React.ReactNode;
}) {
  return (
    <div className="product-card">
      <div className="product-image">{image}</div>
      <div className="product-info">{info}</div>
      <div className="product-actions">{actions}</div>
    </div>
  );
}
```

4-10 & 4-11 HOC 高阶组件

```
// HOC: 高阶组件 (Higher-Order Component)
function withLogger<P extends object>(Component: React.ComponentType<P>) {
  return function withLogger(props: P) {
```



```
useEffect(() => {
  console.log(`${Component.displayName || Component.name} mounted`);
  return () => console.log(`${Component.displayName || Component.name} unmounted`);
}, []);

return <Component {...props} />;
};
}

// HOC: withLoading
function withLoading<P extends { loading?: boolean }>(
  Component: React.ComponentType<P>
) {
  return function WithLoading(props: P) {
    if (props.loading) return <div>Loading...</div>;
    return <Component {...props} />;
  };
}

// 使用 HOC
const ProductCardWithLogging = withLogger(ProductCard);
const ProductCardWithLoading = withLoading(ProductCard);
```

HOC vs Hooks vs Render Props

维度	HOC	Render Props	Hooks
模式	装饰器模式	函数作为 children	组合式函数
命名冲突	⚠️ 容易冲突	✅ 不冲突	✅ 不冲突
嵌套层级	深	深（嵌套地狱）	浅
推荐度	★★	★	★★★★★

第5章：React+Tailwind CSS 原子化样式

5-1 掌握 React 和 Tailwind CSS

Tailwind CSS 是**原子化 CSS** 框架，提供大量工具类（utility class）来快速构建 UI。

5-2 CSS 四大酷刑 vs Tailwind 魔法

传统 CSS 痛点	Tailwind 解决方案
命名困难（BEM/语义化）	不命名，直接用工具类
样式冲突（全局污染）	作用域天然隔离
重复代码多	组合可复用工具类

传统 CSS 痛点	Tailwind 解决方案
难以维护	修改即改 HTML

5-3 React + Tailwind CSS 开发配置

```
npm install -D tailwindcss @tailwindcss/vite
```

```
// vite.config.ts
import { defineConfig } from 'vite';
import react from '@vitejs/plugin-react';
import tailwindcss from '@tailwindcss/vite';

export default defineConfig({
  plugins: [react(), tailwindcss()],
});
```

```
/* index.css */
@import "tailwindcss";
```

5-4 Flex 三板斧

```
<div className="flex justify-center items-center h-screen">
  <div className="flex-1 bg-blue-500 p-4">项目1</div>
  <div className="flex-1 bg-green-500 p-4">项目2</div>
  <div className="flex-1 bg-red-500 p-4">项目3</div>
</div>
```

5-5 Flex 终极对齐

```
justify-content: justify-start | justify-center | justify-end | justify-between | justify-around | justify-evenly
align-items: items-start | items-center | items-end | items-stretch | items-baseline
flex-direction: flex-row | flex-col | flex-row-reverse | flex-col-reverse
```

5-6 Grid 网格布局

```
<div className="grid grid-cols-3 gap-4">
  {products.map(product => (
    <div key={product.id} className="bg-white rounded-lg shadow p-4">
      <h3>{product.name}</h3>
    </div>
  ))}
</div>
```

5-7 手机、平板、PC：三屏联动响应式

```
<div className="
  grid
  grid-cols-1      /* 手机：1列 */
  sm:grid-cols-2   /* 平板：2列 */
  lg:grid-cols-3   /* 桌面：3列 */
  xl:grid-cols-4   /* 大屏：4列 */
  gap-4
  p-4
">
  {products.map(product => (
    <ProductCard key={product.id} product={product} />
  ))}
</div>
```

5-10 状态特效

```
<button className="
  px-6 py-3 bg-blue-600 text-white rounded-lg font-semibold
  hover:bg-blue-700 hover:shadow-lg hover:scale-105
  focus:outline-none focus:ring-2 focus:ring-blue-400
  transition-all duration-200
">
  悬停发光按钮
</button>

<input className="
  w-full px-4 py-2 border border-gray-300 rounded-lg
  focus:border-blue-500 focus:ring-2 focus:ring-blue-200
  outline-none transition-all duration-200
" />
```

5-11 黑暗模式：React + Tailwind CSS

```
// 使用 Tailwind 的 dark: 前缀
function ThemedCard() {
  const [dark, setDark] = useState(false);

  return (
    <div className={dark ? 'dark' : ''}>
      <div className="bg-white dark:bg-gray-800 text-black dark:text-white p-6 rounded-lg shadow">
        <h2 className="text-xl font-bold">暗黑模式切换</h2>
        <p className="mt-2 text-gray-600 dark:text-gray-300">
          使用 Tailwind 的 dark: 前缀实现
        </p>
        <button
          onClick={() => setDark(!dark)}
          className="mt-4 px-4 py-2 bg-blue-500 dark:bg-yellow-500 text-white rounded"
        >
```

```

    >
    切换 {dark ? '亮色' : '暗黑'} 模式
  </button>
</div>
</div>
);
}

```

5-13 主题与指令：自定义 CSS 变量

```

@import "tailwindcss";

@theme {
  --color-primary: #007bff;
  --color-secondary: #6c757d;
  --color-accent: #ffc107;
}

```

5-14 综合实战：产品推荐组件

```

function ProductRecommendations() {
  const products = [
    { id: 1, name: '无线耳机', price: 299, image: '/earbuds.jpg', rating: 4.5 },
    { id: 2, name: '智能手表', price: 1299, image: '/watch.jpg', rating: 4.8 },
  ];

  return (
    <div className="max-w-7xl mx-auto px-4 py-8">
      <h2 className="text-2xl font-bold text-gray-900 dark:text-white mb-6">
        为你推荐
      </h2>
      <div className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-6">
        {products.map(product => (
          <div key={product.id}
            className="bg-white dark:bg-gray-800 rounded-xl shadow-md overflow-hidden
              hover:shadow-xl transition-shadow duration-300">
            <img src={product.image} alt={product.name}
              className="w-full h-48 object-cover" />
            <div className="p-4">
              <h3 className="text-lg font-semibold text-gray-900 dark:text-white">
                {product.name}
              </h3>
              <div className="flex items-center mt-2">
                <span className="text-yellow-400">{'★'.repeat(Math.floor(product.rating))}</span>
                <span className="text-gray-500 ml-1">{product.rating}</span>
              </div>
              <div className="flex items-center justify-between mt-3">
                <span className="text-xl font-bold text-red-500">¥{product.price}</span>
                <button className="px-4 py-2 bg-blue-600 text-white rounded-lg
                  hover:bg-blue-700 transition-colors">

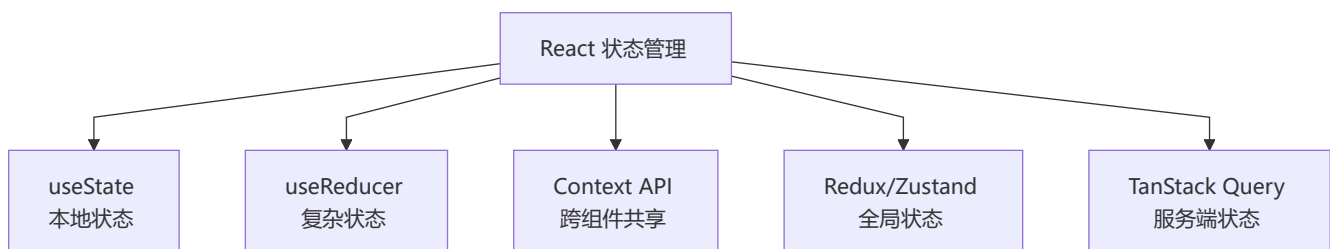
```

```
        加入购物车
      </button>
    </div>
  </div>
</div>
))}
</div>
</div>
);
}
```



第6章：状态管理与组件交互

6-1 React 状态管理核心技术



6-2 从 let 到 useState 的正确打开方式

```
// ❌ 错误：普通变量不会触发重新渲染
function Wrong() {
  let count = 0;
  const increment = () => { count++; };
  return <button onClick={increment}>{count}</button>; // 显示不会更新!
}

// ✅ 正确：useState
function Correct() {
  const [count, setCount] = useState(0);
  return <button onClick={() => setCount(c => c + 1)}>{count}</button>;
}
```

6-3 React State 深度解析

State 特性

- **异步更新**: `setState` 不会立即更新，React 会批量处理
- **快照机制**: 每次渲染都有独立的 state 快照
- **不可变性**: 必须创建新值，不能直接修改现有 state

```
function Counter() {
  const [count, setCount] = useState(0);

  const handleClick = () => {
    setCount(count + 1);
    console.log(count); // 依然是旧的 count 值!
  };

  return <button onClick={handleClick}>{count}</button>;
}
```

6-4 & 6-5 React 状态管理的特点

批处理 (Batching)

```
// React 18+ 自动批处理
function handleClick() {
  setCount(c => c + 1);
  setFlag(f => !f);
  setText('hello');
  // 只触发一次重新渲染
}
```

函数式更新

```
// ✅ 正确：使用函数式更新避免闭包陷阱
setCount(prev => prev + 1);

// 连续多次更新
setCount(prev => prev + 1);
setCount(prev => prev + 1);
setCount(prev => prev + 1);
// 结果: count 增加 3
```

6-6 State 实战：SKU选择器

```
interface SKU {
  color: string;
  size: string;
  stock: number;
  price: number;
}

function SKUSElector({ skus }: { skus: SKU[] }) {
  const [selectedColor, setSelectedColor] = useState('');
  const [selectedSize, setSelectedSize] = useState('');

  const availableSizes = useMemo(() =>
    [...new Set(skus.filter(s => !selectedColor || s.color === selectedColor).map(s =>
      s.size))],
```

```

[skus, selectedColor]
);

const currentSKU = useMemo(() =>
  skus.find(s => s.color === selectedColor && s.size === selectedSize),
  [skus, selectedColor, selectedSize]
);

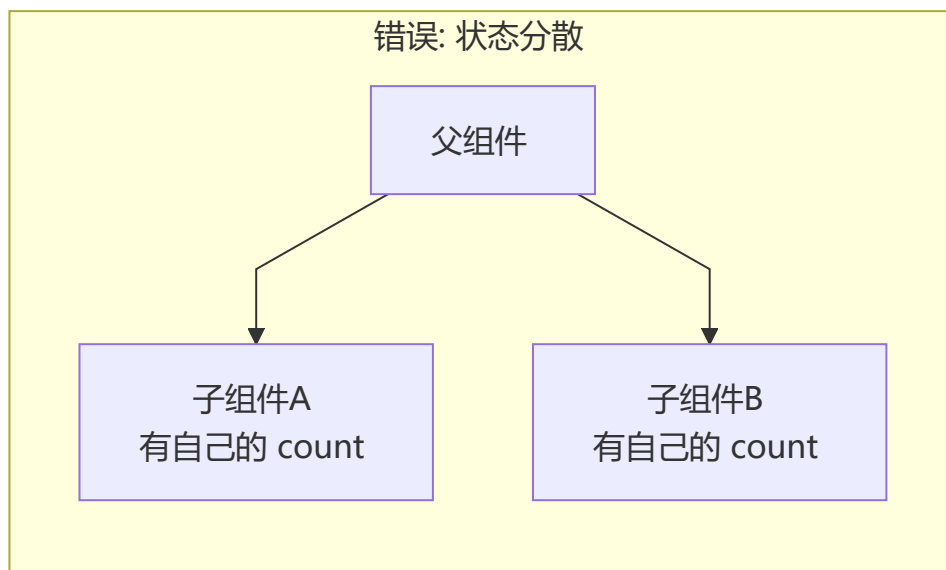
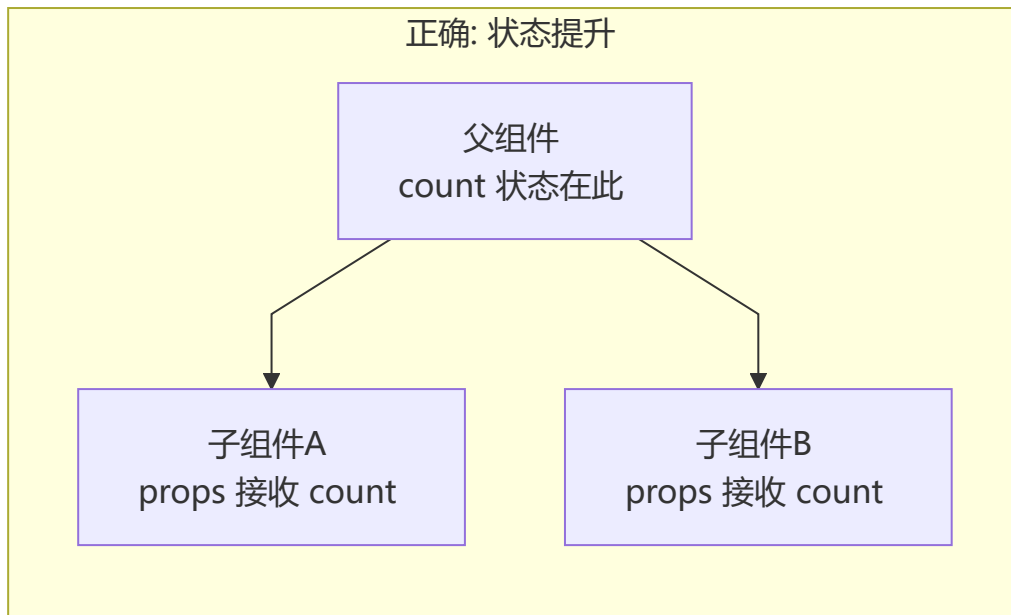
return (
  <div>
    <div className="mb-4">
      <label className="block mb-2">颜色: </label>
      <div className="flex gap-2">
        {[...new Set(skus.map(s => s.color))].map(color => (
          <button key={color}
            onClick={() => { setSelectedColor(color); setSelectedSize(''); }}
            className={`px-4 py-2 rounded ${selectedColor === color ? 'bg-blue-500 text-white' : 'bg-gray-200'}`}>
              {color}
            </button>
          )]}
      </div>
    </div>

    <div className="mb-4">
      <label className="block mb-2">尺寸: </label>
      <div className="flex gap-2">
        {availableSizes.map(size => (
          <button key={size}
            onClick={() => setSelectedSize(size)}
            className={`px-4 py-2 rounded ${selectedSize === size ? 'bg-blue-500 text-white' : 'bg-gray-200'}`}>
              {size}
            </button>
          )]}
      </div>
    </div>

    {currentSKU && (
      <div className="p-4 bg-gray-50 rounded">
        <p>价格: ¥{currentSKU.price}</p>
        <p>库存: {currentSKU.stock > 0 ? `${currentSKU.stock}件` : '已售罄'}</p>
      </div>
    )}
  </div>
);
}

```

6-7 & 6-8 状态提升 (Lifting State Up)



```
function Parent() {  
  const [count, setCount] = useState(0);  
  
  return (  
    <div>  
      <CounterDisplay count={count} />  
      <CounterControls count={count} setCount={setCount} />  
    </div>  
  );  
}  
  
function CounterDisplay({ count }: { count: number }) {  
  return <h2>计数: {count}</h2>;  
}
```



```
function CounterControls({ count, setCount }: {
  count: number;
  setCount: React.Dispatch<React.SetStateAction<number>>;
}) {
  return (
    <div>
      <button onClick={() => setCount(c => c + 1)}>+</button>
      <button onClick={() => setCount(c => c - 1)}>-</button>
    </div>
  );
}
```

6-9 深层嵌套 State 更新避坑

```
interface User {
  name: string;
  address: {
    city: string;
    district: string;
    detail: string;
  };
  hobbies: string[];
}

const [user, setUser] = useState<User>({
  name: '张三',
  address: { city: '北京', district: '海淀', detail: '...' },
  hobbies: ['读书', '跑步'],
});

// ✅ 正确更新深层嵌套
function updateDistrict(district: string) {
  setUser(prev => ({
    ...prev,
    address: {
      ...prev.address,
      district,
    },
  }));
}

// ✅ 更新数组
function addHobby(hobby: string) {
  setUser(prev => ({
    ...prev,
    hobbies: [...prev.hobbies, hobby],
  }));
}
```

6-10 State 不可变性

```
// ❌ 错误: 直接修改 state
const [todos, setTodos] = useState(['a', 'b']);
todos.push('c'); // 直接修改
setTodos(todos); // React 不会检测到变化!

// ✅ 正确: 创建新数组
setTodos([...todos, 'c']);

// ✅ 删除
setTodos(todos.filter(t => t !== 'a'));

// ✅ 修改
setTodos(todos.map(t => t === 'a' ? 'A' : t));
```

6-11 复杂 State 更新的艺术: Immer

```
npm install immer
```

```
import { produce } from 'immer';

const [user, setUser] = useState<User>({ ... });

// 使用 Immer: 以可变的方式写不可变逻辑
function updateAddress(district: string) {
  setUser(produce(draft => {
    draft.address.district = district;
  }));
}

function addHobby(hobby: string) {
  setUser(produce(draft => {
    draft.hobbies.push(hobby);
  }));
}
```

6-12 柯里化 + Immer: 万能 State 更新器

```
function useCurriedUpdate<T extends object>(initialState: T) {
  const [state, setState] = useState(initialState);

  const createUpdate = useCallback(<K extends keyof T>(key: K) => {
    return (value: T[K] | ((prev: T[K]) => T[K])) => {
      setState(produce(draft => {
        draft[key] = typeof value === 'function'
          ? (value as (prev: T[K]) => T[K])(draft[key])
          : value;
      }));
    };
  });
```

```

    }, []));

    return [state, createUpdate] as const;
  }

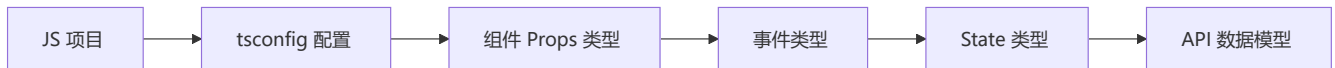
  // 使用
  const [user, update] = useCurriedUpdate(userData);
  const updateName = update('name');
  const updateCity = (city: string) => {
    setUser(produce(draft => {
      draft.address.city = city;
    }));
  };
};

```



第7章：React + TypeScript 类型安全

7-1 类型安全体系与实践路线



7-2 从JS平滑升级到TS

```

// tsconfig.json
{
  "compilerOptions": {
    "strict": true,
    "jsx": "react-jsx",
    "target": "ES2020",
    "module": "ESNext",
    "moduleResolution": "bundler",
    "skipLibCheck": true
  }
}

```

7-3 & 7-4 组件 Props 类型声明

```

// 方式1: type 别名
type ButtonProps = {
  label: string;
  onClick: () => void;
  variant?: 'primary' | 'secondary' | 'danger';
  disabled?: boolean;
  size?: 'sm' | 'md' | 'lg';
};

// 方式2: interface
interface CardProps {
  title: string;
}

```

```

children: React.ReactNode;
footer?: React.ReactNode;
className?: string;
}

```

7-5 interface + 联合类型封装万能按钮

```

type ButtonVariant = 'primary' | 'secondary' | 'danger' | 'ghost';
type ButtonSize = 'sm' | 'md' | 'lg';

interface ButtonProps {
  variant?: ButtonVariant;
  size?: ButtonSize;
  disabled?: boolean;
  loading?: boolean;
  children: React.ReactNode;
  onClick?: () => void;
  type?: 'button' | 'submit' | 'reset';
}

function Button({
  variant = 'primary',
  size = 'md',
  disabled = false,
  loading = false,
  children,
  onClick,
  type = 'button',
}: ButtonProps) {
  const baseClass = 'px-4 py-2 rounded font-semibold transition-all duration-200';
  const variantClass = {
    primary: 'bg-blue-600 text-white hover:bg-blue-700',
    secondary: 'bg-gray-200 text-gray-800 hover:bg-gray-300',
    danger: 'bg-red-600 text-white hover:bg-red-700',
    ghost: 'bg-transparent text-gray-600 hover:bg-gray-100',
  }[variant];

  return (
    <button
      type={type}
      onClick={onClick}
      disabled={disabled || loading}
      className={` ${baseClass} ${variantClass} ${disabled ? 'opacity-50 cursor-not-allowed'
: ''} `}
      >
      {loading ? '加载中...' : children}
    </button>
  );
}

```

7-6 interface extends 揭秘组件继承

```
interface BaseInputProps {
  label: string;
  error?: string;
  required?: boolean;
  disabled?: boolean;
}

interface TextInputProps extends BaseInputProps {
  type?: 'text' | 'email' | 'password';
  placeholder?: string;
  value: string;
  onChange: (value: string) => void;
}

interface NumberInputProps extends BaseInputProps {
  min?: number;
  max?: number;
  step?: number;
  value: number;
  onChange: (value: number) => void;
}
```

7-7 .d.ts 到底是干啥的?

```
// src/types/global.d.ts
declare module '*.module.css' {
  const classes: { readonly [key: string]: string };
  export default classes;
}

declare module '*.svg' {
  const content: string;
  export default content;
}

// 为环境变量添加类型
/// <reference types="vite/client" />
interface ImportMetaEnv {
  readonly VITE_API_URL: string;
  readonly VITE_APP_TITLE: string;
}
```

7-8 React 事件类型

```
// 鼠标事件
function handleClick(e: React.MouseEvent<HTMLButtonElement>) {
  console.log(e.clientX, e.clientY);
}
```

```
// 表单事件
function handleChange(e: React.ChangeEvent<HTMLInputElement>) {
  console.log(e.target.value);
}

// 表单提交
function handleSubmit(e: React.FormEvent<HTMLFormElement>) {
  e.preventDefault();
  console.log('submitted');
}

// 键盘事件
function handleKeyDown(e: React.KeyboardEvent<HTMLInputElement>) {
  if (e.key === 'Enter') console.log('pressed enter');
}
```

7-10 useState 类型安全

```
// 类型推断
const [count, setCount] = useState(0); // type: number

// 显式类型
const [user, setUser] = useState<User | null>(null);

// 联合类型
type Status = 'idle' | 'loading' | 'success' | 'error';
const [status, setStatus] = useState<Status>('idle');
```

7-11 TS + useState 构建强类型电商状态

```
interface CartItem {
  id: number;
  productId: number;
  name: string;
  price: number;
  quantity: number;
  image: string;
  sku: string;
}

interface CartState {
  items: CartItem[];
  totalAmount: number;
  totalCount: number;
  coupon?: string;
  discount?: number;
}

function useCart() {
  const [cart, setCart] = useState<CartState>({
```

```

    items: [],
    totalAmount: 0,
    totalCount: 0,
  });

  const addItem = useCallback((item: Omit<CartItem, 'quantity'>) => {
    setCart(prev => {
      const existing = prev.items.find(i => i.sku === item.sku);
      if (existing) {
        return {
          ...prev,
          items: prev.items.map(i =>
            i.sku === item.sku ? { ...i, quantity: i.quantity + 1 } : i
          ),
          totalCount: prev.totalCount + 1,
          totalAmount: prev.totalAmount + item.price,
        };
      }
      return {
        ...prev,
        items: [...prev.items, { ...item, quantity: 1 }],
        totalCount: prev.totalCount + 1,
        totalAmount: prev.totalAmount + item.price,
      };
    });
  }, []);

  return { cart, addItem };
}

```



第8章：React Router v6 路由系统

8-1 React Router 是什么？

React Router 是 React 生态中最流行的路由库，实现 SPA 的**客户端路由**，在不刷新页面的情况下切换视图。

8-2 BrowserRouter + Route

```
npm install react-router-dom
```

```

import { BrowserRouter, Routes, Route } from 'react-router-dom';
import Home from './pages/Home';
import About from './pages/About';
import Product from './pages/Product';

function App() {
  return (
    <BrowserRouter>
      <Routes>
        <Route path="/" element={<Home />} />

```

```

    <Route path="/about" element={<About />} />
    <Route path="/product/:id" element={<Product />} />
    <Route path="*" element={<NotFound />} />
  </Routes>
</BrowserRouter>
);
}

```

8-3 index 文件的妙用

```

<Route path="/dashboard" element={<DashboardLayout />}>
  <Route index element={<DashboardHome />} />
  <Route path="settings" element={<Settings />} />
  <Route path="analytics" element={<Analytics />} />
</Route>

```

`index` 路由匹配父路径的精确路径 `/dashboard`。

8-4 Outlet：专业级路由布局

```

import { Outlet } from 'react-router-dom';

function DashboardLayout() {
  return (
    <div className="dashboard-layout">
      <aside className="sidebar">
        <nav>{/* 导航菜单 */}</nav>
      </aside>
      <main className="content">
        <Outlet /> {/* 子路由在此渲染 */}
      </main>
    </div>
  );
}

```

8-5 createBrowserRouter 全解析

```

import { createBrowserRouter, RouterProvider } from 'react-router-dom';

const router = createBrowserRouter([
  {
    path: '/',
    element: <RootLayout />,
    errorElement: <ErrorPage />,
    children: [
      { index: true, element: <Home /> },
      {
        path: 'products',
        element: <ProductLayout />,
        children: [

```



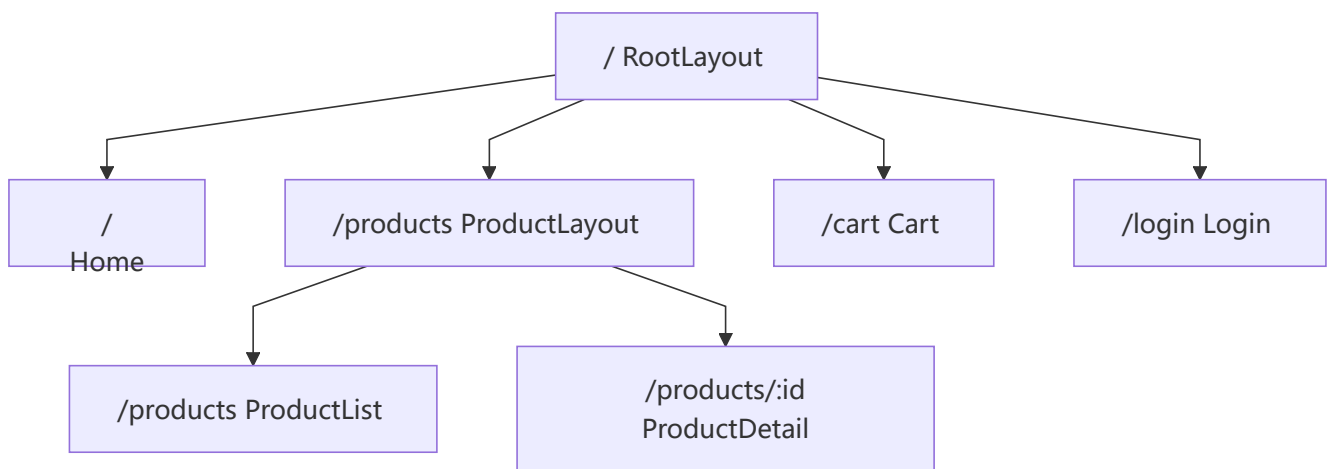
```

    { index: true, element: <ProductList /> },
    { path: ':id', element: <ProductDetail /> },
  ],
},
{ path: 'cart', element: <Cart /> },
{ path: 'login', element: <Login /> },
{ path: '*', element: <NotFound /> },
],
},
]);

function App() {
  return <RouterProvider router={router} />;
}

```

8-6 中大型项目：createBrowserRouter + Layout



8-7 404 + Error Boundary

```

// 404 路由
{ path: '*', element: <NotFound /> }

// Error Boundary
import { useRouteError } from 'react-router-dom';

function ErrorPage() {
  const error = useRouteError() as { statusText?: string; message?: string };
  return (
    <div className="error-page">
      <h1>出错了! </h1>
      <p>{error.statusText || error.message}</p>
    </div>
  );
}

```

8-8 Link vs NavLink

```
import { Link, NavLink } from 'react-router-dom';

// Link: 基本导航
<Link to="/about">关于我们</Link>

// NavLink: 带激活状态的导航
<NavLink
  to="/products"
  className={({ isActive }) => isActive ? 'nav-link active' : 'nav-link'}
  style={({ isActive }) => ({ fontWeight: isActive ? 'bold' : 'normal' })}
>
  产品列表
</NavLink>
```

8-9 程式化导航: useNavigate

```
import { useNavigate } from 'react-router-dom';

function LoginButton() {
  const navigate = useNavigate();

  const handleLogin = async () => {
    await login();
    navigate('/dashboard', { replace: true }); // 替换历史记录
  };

  return <button onClick={handleLogin}>登录</button>;
}

// 返回上一页
function BackButton() {
  const navigate = useNavigate();
  return <button onClick={() => navigate(-1)}>返回</button>;
}
```

8-10 URL 参数匹配

```
// useParams: 路径参数
// 路由: /product/:id
import { useParams } from 'react-router-dom';

function ProductDetail() {
  const { id } = useParams<{ id: string }>();
  // 使用 id 获取产品数据
  return <div>产品ID: {id}</div>;
}

// useSearchParams: 查询参数
```

```
// URL: /products?category=electronics&sort=price
function ProductList() {
  const [searchParams, setSearchParams] = useSearchParams();
  const category = searchParams.get('category') || 'all';

  const updateFilter = (key: string, value: string) => {
    setSearchParams(prev => {
      prev.set(key, value);
      return prev;
    });
  };

  return (
    <div>
      <select value={category} onChange={e => updateFilter('category', e.target.value)}>
        <option value="all">全部分类</option>
        <option value="electronics">电子产品</option>
        <option value="clothing">服装</option>
      </select>
    </div>
  );
}
```

8-13 鉴权集成

```
// 路由守卫
function ProtectedRoute({ children }: { children: React.ReactNode }) {
  const isAuthenticated = useAuth();

  if (!isAuthenticated) {
    return <Navigate to="/login" replace />;
  }

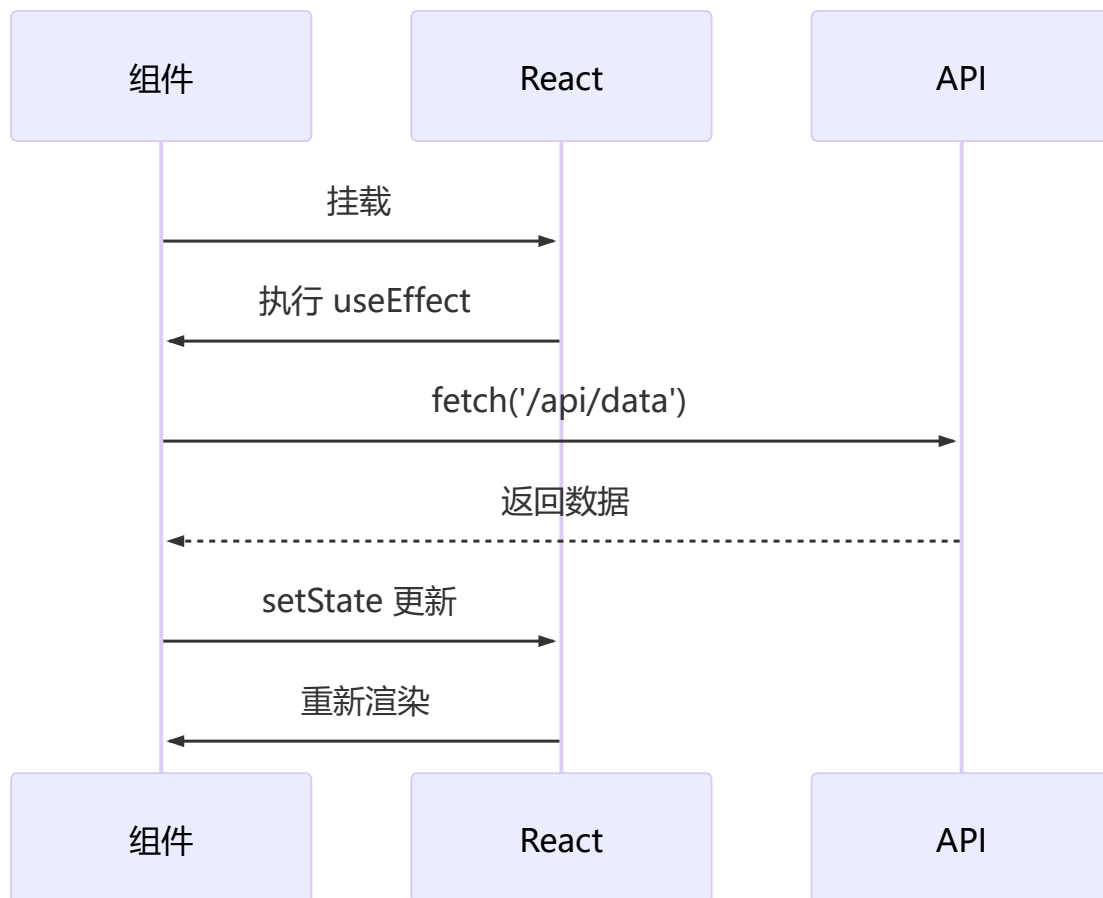
  return <>{children}</>;
}

// 路由配置
const router = createBrowserRouter([
  {
    path: '/dashboard',
    element: (
      <ProtectedRoute>
        <DashboardLayout />
      </ProtectedRoute>
    ),
    children: [
      { index: true, element: <DashboardHome /> },
    ],
  },
  { path: '/login', element: <Login /> },
]);
```



第9章：副作用处理 useEffect 与异步请求

9-2 异步请求的本质



9-3 useEffect 详解

```
useEffect(() => {
  // 副作用逻辑
  return () => {
    // 清理函数（组件卸载或依赖变化时执行）
  };
}, [dependencies]);
```

三种执行时机

```
// 1. 每次渲染后执行
useEffect(() => {
  console.log('每次渲染后执行');
});

// 2. 仅挂载时执行
useEffect(() => {
  console.log('仅在挂载时执行');
  return () => console.log('卸载时清理');
});
```

```
}, []);
```

// 3. 依赖变化时执行

```
useEffect(() => {  
  console.log('count 或 name 变化时执行');  
}, [count, name]);
```

9-4 用 useEffect 搜索功能

```
function SearchProducts() {  
  const [query, setQuery] = useState('');  
  const [results, setResults] = useState<Product[]>([]);  
  const [loading, setLoading] = useState(false);  
  
  useEffect(() => {  
    if (!query) {  
      setResults([]);  
      return;  
    }  
  
    const fetchProducts = async () => {  
      setLoading(true);  
      try {  
        const res = await fetch(`/api/products?q=${query}`);  
        const data = await res.json();  
        setResults(data);  
      } catch (err) {  
        console.error('搜索失败', err);  
      } finally {  
        setLoading(false);  
      }  
    };  
  
    fetchProducts();  
  }, [query]);  
  
  return (  
    <div>  
      <input value={query} onChange={e => setQuery(e.target.value)} placeholder="搜索产品..." />  
      {loading && <div>搜索中...</div>}  
      <ul>  
        {results.map(product => (  
          <li key={product.id}>{product.name}</li>  
        ))}  
      </ul>  
    </div>  
  );  
}
```

9-5 错误处理策略

```
interface AsyncState<T> {
  data: T | null;
  loading: boolean;
  error: string | null;
}

function useProducts() {
  const [state, setState] = useState<AsyncState<Product[]>>({
    data: null,
    loading: true,
    error: null,
  });

  useEffect(() => {
    const fetchProducts = async () => {
      try {
        setState(prev => ({ ...prev, loading: true, error: null }));
        const res = await fetch('/api/products');
        if (!res.ok) throw new Error(`HTTP ${res.status}`);
        const data = await res.json();
        setState({ data, loading: false, error: null });
      } catch (err) {
        setState({
          data: null,
          loading: false,
          error: err instanceof Error ? err.message : '未知错误',
        });
      }
    };

    fetchProducts();
  }, []);

  return state;
}
```

9-6 依赖数组机制

```
// ✅ 依赖数组中的值变化时重新执行
useEffect(() => {
  document.title = `${count} 条消息`;
}, [count]);

// ❌ 缺少依赖：闭包陷阱
useEffect(() => {
  const timer = setInterval(() => {
    console.log(count); // 永远是初始值
  }, 1000);
  return () => clearInterval(timer);
}, []); // 缺少 count
```

9-7 清理函数：内存泄漏不再

```
useEffect(() => {
  const handleResize = () => console.log(window.innerWidth);
  window.addEventListener('resize', handleResize);

  // 清理函数在卸载时执行
  return () => {
    window.removeEventListener('resize', handleResize);
  };
}, []);
```

9-8 AbortController：API 请求的竞争与取消

```
useEffect(() => {
  const controller = new AbortController();

  const fetchData = async () => {
    try {
      const res = await fetch('/api/products', {
        signal: controller.signal,
      });
      const data = await res.json();
      setProducts(data);
    } catch (err) {
      if (err instanceof DOMException && err.name === 'AbortError') {
        console.log('请求已取消');
        return;
      }
      console.error('请求失败', err);
    }
  };

  fetchData();
}, []);
```

```

return () => {
  controller.abort(); // 组件卸载时取消请求
};
}, []);

```

9-9 输入防抖: useDebounce

```

function useDebounce<T>(value: T, delay: number): T {
  const [debouncedValue, setDebouncedValue] = useState(value);

  useEffect(() => {
    const timer = setTimeout(() => {
      setDebouncedValue(value);
    }, delay);

    return () => clearTimeout(timer);
  }, [value, delay]);

  return debouncedValue;
}

// 使用
function SearchComponent() {
  const [query, setQuery] = useState('');
  const debouncedQuery = useDebounce(query, 500);

  useEffect(() => {
    if (debouncedQuery) {
      // 只在用户停止输入 500ms 后发起请求
      searchAPI(debouncedQuery);
    }
  }, [debouncedQuery]);

  return <input value={query} onChange={e => setQuery(e.target.value)} />;
}

```

9-12 React Router Loaders

```

import { createBrowserRouter } from 'react-router-dom';

const router = createBrowserRouter([
  {
    path: '/products/:id',
    element: <ProductDetail />,
    loader: async ({ params }) => {
      const res = await fetch(`/api/products/${params.id}`);
      if (!res.ok) throw new Response('Not Found', { status: 404 });
      return res.json();
    },
    errorElement: <ErrorPage />,
  },
]);

```



```
  ]);

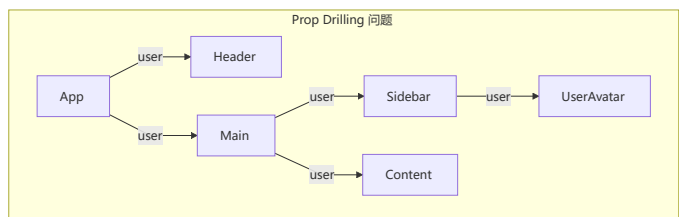
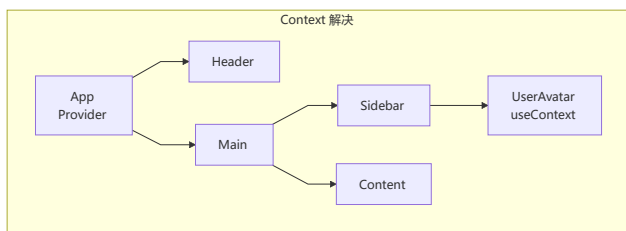
// 组件中获取数据
import { useLoaderData } from 'react-router-dom';

function ProductDetail() {
  const product = useLoaderData() as Product;
  return <div>{product.name}</div>;
}
```

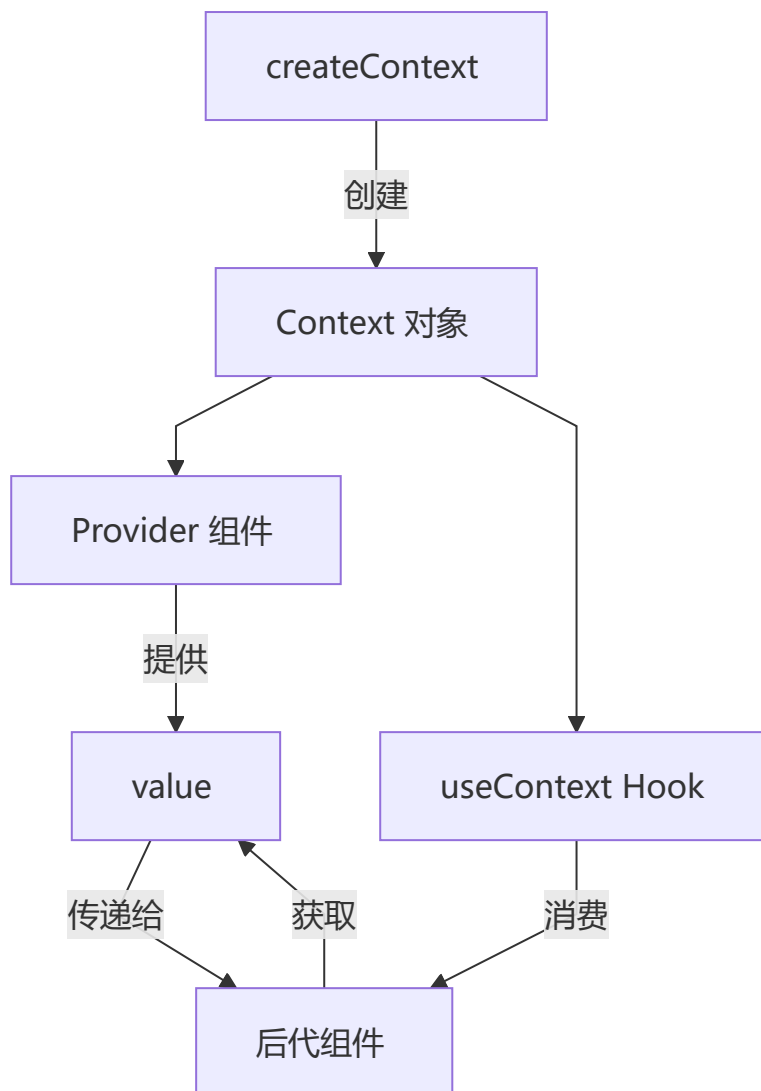


第10章：Context API 跨组件状态管理

10-2 状态提升破解 Prop Drilling



10-3 Context API 核心原理



10-4 Context 创建：购物车状态

```
interface CartItem {
  id: number;
  name: string;
  price: number;
  quantity: number;
  image: string;
}

interface CartContextType {
  items: CartItem[];
  addItem: (item: CartItem) => void;
  removeItem: (id: number) => void;
  updateQuantity: (id: number, quantity: number) => void;
  clearCart: () => void;
  totalAmount: number;
  totalCount: number;
}
```

```

const CartContext = createContext<CartItem[] | undefined>(undefined);

export function CartProvider({ children }: { children: React.ReactNode }) {
  const [items, setItems] = useState<CartItem[]>([]);

  const addItem = useCallback((item: CartItem) => {
    setItems(prev => {
      const existing = prev.find(i => i.id === item.id);
      if (existing) {
        return prev.map(i =>
          i.id === item.id ? { ...i, quantity: i.quantity + item.quantity } : i
        );
      }
      return [...prev, item];
    });
  }, []);

  const totalAmount = useMemo(() =>
    items.reduce((sum, item) => sum + item.price * item.quantity, 0),
    [items]
  );

  const totalCount = useMemo(() =>
    items.reduce((sum, item) => sum + item.quantity, 0),
    [items]
  );

  return (
    <CartContext.Provider value={{
      items, addItem,
      removeItem: (id) => setItems(prev => prev.filter(i => i.id !== id)),
      updateQuantity: (id, qty) => setItems(prev =>
        prev.map(i => i.id === id ? { ...i, quantity: qty } : i)
      ),
      clearCart: () => setItems([]),
      totalAmount, totalCount,
    }}>
      {children}
    </CartContext.Provider>
  );
}

export function useCart() {
  const context = useContext(CartContext);
  if (!context) throw new Error('useCart must be used within CartProvider');
  return context;
}

```

10-5 Context 订阅与发布

```
// 根组件包裹
```

```
function App() {
  return (
    <CartProvider>
      <Header />
      <MainContent />
    </CartProvider>
  );
}

// 任意后代组件消费
function CartBadge() {
  const { totalCount } = useCart();
  return (
    <span className="cart-badge">
      购物车 ({totalCount})
    </span>
  );
}
```

10-6 购物车 UI 重构

```
function CartPage() {
  const { items, updateQuantity, removeItem, totalAmount } = useCart();

  return (
    <div className="max-w-4xl mx-auto p-6">
      <h1 className="text-2xl font-bold mb-6">购物车</h1>

      {items.length === 0 ? (
        <EmptyCart />
      ) : (
        <>
          <div className="space-y-4">
            {items.map(item => (
              <div key={item.id} className="flex items-center gap-4 p-4 bg-white rounded-lg shadow">
                <img src={item.image} alt={item.name} className="w-20 h-20 object-cover rounded" />
                <div className="flex-1">
                  <h3 className="font-semibold">{item.name}</h3>
                  <p className="text-gray-500">¥{item.price}</p>
                </div>
                <div className="flex items-center gap-2">
                  <button onClick={() => updateQuantity(item.id, item.quantity - 1)}
                    className="px-2 py-1 border rounded">-</button>
                  <span className="w-8 text-center">{item.quantity}</span>
                  <button onClick={() => updateQuantity(item.id, item.quantity + 1)}
                    className="px-2 py-1 border rounded">+</button>
                </div>
                <p className="font-bold">¥{(item.price * item.quantity).toFixed(2)}</p>
                <button onClick={() => removeItem(item.id)}
                  className="text-red-500 hover:text-red-700">删除</button>
              </div>
            )})
          </div>
        </>
      )}
    </div>
  );
}
```

```

        </div>
      )}
    </div>

    <div className="mt-6 p-4 bg-gray-50 rounded-lg text-right">
      <p className="text-lg">合计: <span className="font-bold text-xl">¥
{totalAmount.toFixed(2)}</span></p>
      <button className="mt-2 px-6 py-2 bg-blue-600 text-white rounded-lg hover:bg-
blue-700">
        去结算
      </button>
    </div>
  </>
})
</div>
);
}

```

10-9 购物车持久化 (localStorage)

```

export function CartProvider({ children }: { children: React.ReactNode }) {
  const [items, setItems] = useState<CartItem[]>(() => {
    try {
      const saved = localStorage.getItem('cart');
      return saved ? JSON.parse(saved) : [];
    } catch {
      return [];
    }
  });

  // 自动持久化
  useEffect(() => {
    localStorage.setItem('cart', JSON.stringify(items));
  }, [items]);

  // 多标签同步
  useEffect(() => {
    const handleStorage = (e: StorageEvent) => {
      if (e.key === 'cart' && e.newValue) {
        setItems(JSON.parse(e.newValue));
      }
    };
    window.addEventListener('storage', handleStorage);
    return () => window.removeEventListener('storage', handleStorage);
  }, []);

  // ... rest of the provider
}

```



第11章：React Hooks 深度赋能

11-1 React Hooks 核心规则

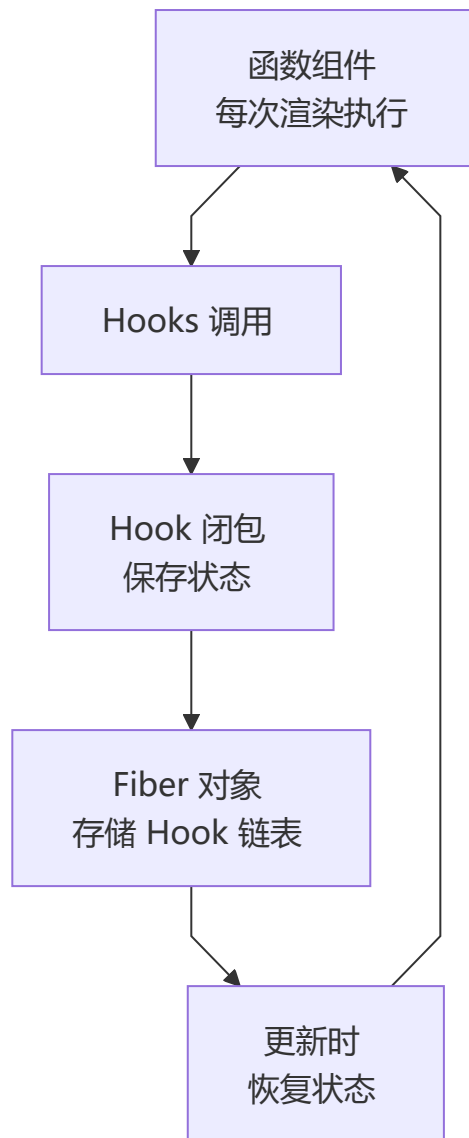
Hooks 三大铁律

1. **只在顶层调用**：不在循环、条件、嵌套函数中调用
2. **只在函数组件或自定义 Hook 中调用**
3. **保证调用顺序不变**：每次渲染执行顺序必须一致

```
// ❌ 错误：条件中调用 hooks
function Bad({ flag }) {
  if (flag) {
    const [count, setCount] = useState(0); // 违反规则！
  }
}

// ✅ 正确
function Good({ flag }) {
  const [count, setCount] = useState(0); // 总是在顶层
}
```

Hooks 工作原理



11-2 闭包冻结 (Stale Closure) 风险

```
function StaleClosureExample() {
  const [count, setCount] = useState(0);

  // ❌ 闭包陷阱
  useEffect(() => {
    const timer = setInterval(() => {
      console.log(count); // 永远是 0!
      setCount(count + 1); // 永远是 1
    }, 1000);
    return () => clearInterval(timer);
  }, []); // 空依赖, count 被冻结

  // ✅ 使用函数式更新
  useEffect(() => {
    const timer = setInterval(() => {
      setCount(prev => prev + 1); // 正确的做法
    }, 1000);
    return () => clearInterval(timer);
  }, []);
```

```

    }, 1000);
    return () => clearInterval(timer);
  }, []);
}

```

11-3 StrictMode 双重调用

React 严格模式下，开发环境的 `useEffect` 会执行两次，用于检测副作用的清理是否正确。

```

function App() {
  return (
    <StrictMode>
      <Main />
    </StrictMode>
  );
}

```

// 开发环境：组件挂载 → 卸载 → 重新挂载
 // 用于检测：清理函数是否正确、是否有内存泄漏

11-4 useCallbakc + React.memo

```

import { memo, useCallback } from 'react';

// React.memo: props 不变时跳过渲染
const ProductCard = memo(function ProductCard({
  product,
  onAddToCart,
}: {
  product: Product;
  onAddToCart: (id: number) => void;
}) {
  console.log('渲染: ', product.name);
  return (
    <div>
      <h3>{product.name}</h3>
      <button onClick={() => onAddToCart(product.id)}>加入购物车</button>
    </div>
  );
});

function ProductList({ products }: { products: Product[] }) {
  // useCallback: 缓存函数引用
  const handleAddToCart = useCallback((id: number) => {
    console.log('加入购物车', id);
  }, []);

  return (
    <div>
      {products.map(product => (
        <ProductCard

```



```

        key={product.id}
        product={product}
        onAddToCart={handleAddToCart}
      />
    )}
  </div>
);
}

```

11-5 useMemo 实战应用

```

function ProductList({ products, searchQuery, category, sortBy }: {
  products: Product[];
  searchQuery: string;
  category: string;
  sortBy: 'price' | 'name' | 'rating';
}) {
  // useMemo: 缓存昂贵计算
  const filteredAndSorted = useMemo(() => {
    console.log('重新计算过滤排序...');
    return products
      .filter(p =>
        (category === 'all' || p.category === category) &&
        p.name.toLowerCase().includes(searchQuery.toLowerCase())
      )
      .sort((a, b) => {
        if (sortBy === 'price') return a.price - b.price;
        if (sortBy === 'name') return a.name.localeCompare(b.name);
        return b.rating - a.rating;
      });
  }, [products, searchQuery, category, sortBy]);

  return (
    <div className="grid grid-cols-3 gap-4">
      {filteredAndSorted.map(product => (
        <ProductCard key={product.id} product={product} />
      ))}
    </div>
  );
}

```

11-6 useRef: 引用持久化与 DOM 操作

```

// 访问 DOM 元素
function AutoFocusInput() {
  const inputRef = useRef<HTMLInputElement>(null);

  useEffect(() => {
    inputRef.current?.focus();
  }, []);
}

```

```

    return <input ref={inputRef} className="border p-2 rounded" />;
  }

  // 保存可变值（不触发重新渲染）
  function Timer() {
    const [count, setCount] = useState(0);
    const intervalRef = useRef<number | null>(null);

    const start = () => {
      intervalRef.current = window.setInterval(() => {
        setCount(c => c + 1);
      }, 1000);
    };

    const stop = () => {
      if (intervalRef.current) {
        clearInterval(intervalRef.current);
        intervalRef.current = null;
      }
    };

    useEffect(() => {
      return () => stop(); // 清理
    }, []);

    return <div>{count}<button onClick={start}>开始</button><button onClick={stop}>停止
  </button></div>;
  }

```

11-7 自定义 Hook: useApiData

```

interface UseApiDataResult<T> {
  data: T | null;
  loading: boolean;
  error: string | null;
  refetch: () => void;
}

function useApiData<T>(url: string): UseApiDataResult<T> {
  const [data, setData] = useState<T | null>(null);
  const [loading, setLoading] = useState(true);
  const [error, setError] = useState<string | null>(null);
  const [trigger, setTrigger] = useState(0);

  useEffect(() => {
    let cancelled = false;
    const controller = new AbortController();

    const fetchData = async () => {
      setLoading(true);
      setError(null);
      try {

```

```

    const res = await fetch(url, { signal: controller.signal });
    if (!res.ok) throw new Error(`HTTP ${res.status}`);
    const json = await res.json();
    if (!cancelled) {
      setData(json);
      setLoading(false);
    }
  } catch (err) {
    if (!cancelled && err instanceof DOMException && err.name !== 'AbortError') {
      setError(err instanceof Error ? err.message : '未知错误');
      setLoading(false);
    }
  }
};

fetchData();

return () => {
  cancelled = true;
  controller.abort();
};
}, [url, trigger]);

const refetch = useCallback(() => {
  setTrigger(t => t + 1);
}, []);

return { data, loading, error, refetch };
}

// 使用
function ProductList() {
  const { data: products, loading, error, refetch } = useApiData<Product[]>
    ('/api/products');

  if (loading) return <Loading />;
  if (error) return <Error message={error} onRetry={refetch} />;
  return <ProductGrid products={products || []} />;
}

```

11-8 自定义 Hook: useLocalStorage

```

function useLocalStorage<T>(key: string, initialValue: T): [T, (value: T | ((prev: T) => T))
=> void] {
  const [storedValue, setStoredValue] = useState<T>(() => {
    try {
      const item = window.localStorage.getItem(key);
      return item ? JSON.parse(item) : initialValue;
    } catch {
      return initialValue;
    }
  });
};

```

```

const setValue = (value: T | ((prev: T) => T)) => {
  try {
    const valueToStore = value instanceof Function ? value(storedValue) : value;
    setStoredValue(valueToStore);
    window.localStorage.setItem(key, JSON.stringify(valueToStore));
  } catch (error) {
    console.error('localStorage 写入失败', error);
  }
};

// 多标签同步
useEffect(() => {
  const handleStorageChange = (e: StorageEvent) => {
    if (e.key === key && e.newValue) {
      setStoredValue(JSON.parse(e.newValue));
    }
  };
  window.addEventListener('storage', handleStorageChange);
  return () => window.removeEventListener('storage', handleStorageChange);
}, [key]);

return [storedValue, setValue];
}

```

第12章：useReducer 复杂状态管理

12-2 useReducer 降维打击复杂场景

```

type CartAction =
  | { type: 'ADD_ITEM'; payload: Product }
  | { type: 'REMOVE_ITEM'; payload: number }
  | { type: 'UPDATE_QUANTITY'; payload: { id: number; quantity: number } }
  | { type: 'APPLY_COUPON'; payload: string }
  | { type: 'CLEAR_CART' };

interface CartState {
  items: CartItem[];
  coupon: string | null;
  discount: number;
}

function cartReducer(state: CartState, action: CartAction): CartState {
  switch (action.type) {
    case 'ADD_ITEM': {
      const existing = state.items.find(i => i.id === action.payload.id);
      if (existing) {
        return {
          ...state,
          items: state.items.map(i =>

```

```

        i.id === action.payload.id
        ? { ...i, quantity: i.quantity + 1 }
        : i
    ),
  };
}
return {
  ...state,
  items: [...state.items, { ...action.payload, quantity: 1 }],
};
}
case 'REMOVE_ITEM':
  return {
    ...state,
    items: state.items.filter(i => i.id !== action.payload),
  };
case 'UPDATE_QUANTITY':
  return {
    ...state,
    items: state.items.map(i =>
      i.id === action.payload.id
        ? { ...i, quantity: Math.max(0, action.payload.quantity) }
        : i
    ),
  };
case 'APPLY_COUPON':
  return { ...state, coupon: action.payload, discount: 0.1 };
case 'CLEAR_CART':
  return { items: [], coupon: null, discount: 0 };
default:
  return state;
}
}

```

12-3 购物车重构：useReducer 四步法

```

function Cart() {
  const [state, dispatch] = useReducer(cartReducer, {
    items: [],
    coupon: null,
    discount: 0,
  });

  const totalAmount = useMemo(() =>
    state.items.reduce((sum, i) => sum + i.price * i.quantity, 0) * (1 - state.discount),
    [state.items, state.discount]
  );

  return (
    <div>
      {state.items.map(item => (
        <CartItem

```

```

        key={item.id}
        item={item}
        onIncrease={() => dispatch({ type: 'UPDATE_QUANTITY', payload: { id: item.id,
quantity: item.quantity + 1 } })}
        onDecrease={() => dispatch({ type: 'UPDATE_QUANTITY', payload: { id: item.id,
quantity: item.quantity - 1 } })}
        onRemove={() => dispatch({ type: 'REMOVE_ITEM', payload: item.id })}
      />
    ))}
    <div>总计: ¥{totalAmount.toFixed(2)}</div>
  </div>
);
}

```

12-5 Action Creator

```

// Action Creator: 统一创建 action
const addItem = (product: Product): CartAction => ({
  type: 'ADD_ITEM',
  payload: product,
});

const removeItem = (id: number): CartAction => ({
  type: 'REMOVE_ITEM',
  payload: id,
});

const updateQuantity = (id: number, quantity: number): CartAction => ({
  type: 'UPDATE_QUANTITY',
  payload: { id, quantity },
});

// 使用
dispatch(addItem(product));
dispatch(removeItem(id));

```

12-6 useReducer + Immer 草案模式

```

import { useImmerReducer } from 'use-immer';

function cartReducer(draft: CartState, action: CartAction) {
  switch (action.type) {
    case 'ADD_ITEM': {
      const existing = draft.items.find(i => i.id === action.payload.id);
      if (existing) {
        existing.quantity += 1;
      } else {
        draft.items.push({ ...action.payload, quantity: 1 });
      }
      break;
    }
  }
}

```

```
case 'REMOVE_ITEM':  
  draft.items = draft.items.filter(i => i.id !== action.payload);  
  break;  
case 'UPDATE_QUANTITY':  
  const item = draft.items.find(i => i.id === action.payload.id);  
  if (item) item.quantity = action.payload.quantity;  
  break;  
case 'APPLY_COUPON':  
  draft.coupon = action.payload;  
  draft.discount = 0.1;  
  break;  
case 'CLEAR_CART':  
  draft.items = [];  
  draft.coupon = null;  
  draft.discount = 0;  
  break;  
}  
}
```



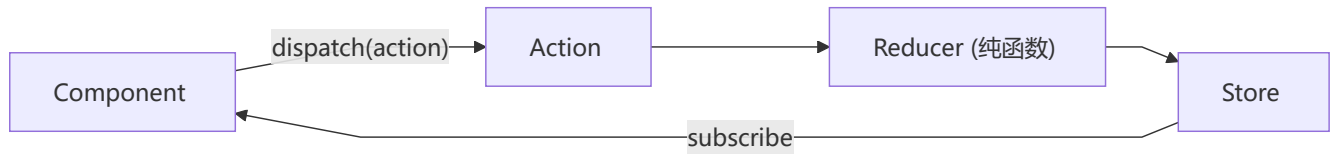
第13章：Redux 生态与 RTK

13-2 Redux 管理状态的核心逻辑

Redux 三大原则



数据流



13-3 Store 构建

```
import { createStore } from 'redux';

interface AppState {
  locale: string;
  theme: 'light' | 'dark';
}

const initialState: AppState = {
  locale: 'zh-CN',
  theme: 'light',
};

function rootReducer(state = initialState, action: any): AppState {
  switch (action.type) {
    case 'SET_LOCALE':
      return { ...state, locale: action.payload };
    case 'SET_THEME':
      return { ...state, theme: action.payload };
    default:
      return state;
  }
}

const store = createStore(rootReducer);
export type RootState = ReturnType<typeof store.getState>;
```


13-7 React-Redux

```
import { useSelector, useDispatch } from 'react-redux';
import type { RootState } from './store';

function ThemeToggle() {
  const theme = useSelector((state: RootState) => state.theme);
  const dispatch = useDispatch();

  return (
    <button onClick={() => dispatch({ type: 'SET_THEME', payload: theme === 'light' ? 'dark' : 'light' })}>
      切换主题
    </button>
  );
}
```

13-8 Redux Toolkit: Slice 架构

```
import { createSlice, configureStore } from '@reduxjs/toolkit';

const cartSlice = createSlice({
  name: 'cart',
  initialState: { items: [] as CartItem[] },
  reducers: {
    addItem: (state, action) => {
      const existing = state.items.find(i => i.id === action.payload.id);
      if (existing) {
        existing.quantity += action.payload.quantity || 1;
      } else {
        state.items.push(action.payload);
      }
    },
    removeItem: (state, action) => {
      state.items = state.items.filter(i => i.id !== action.payload);
    },
    clearCart: (state) => {
      state.items = [];
    },
  },
});

export const { addItem, removeItem, clearCart } = cartSlice.actions;

const store = configureStore({
  reducer: {
    cart: cartSlice.reducer,
    // 其他 slice...
  },
});
```

```
export type RootState = ReturnType<typeof store.getState>;
export type AppDispatch = typeof store.dispatch;
```

13-9 & 13-10 RTK 异步管理: createAsyncThunk

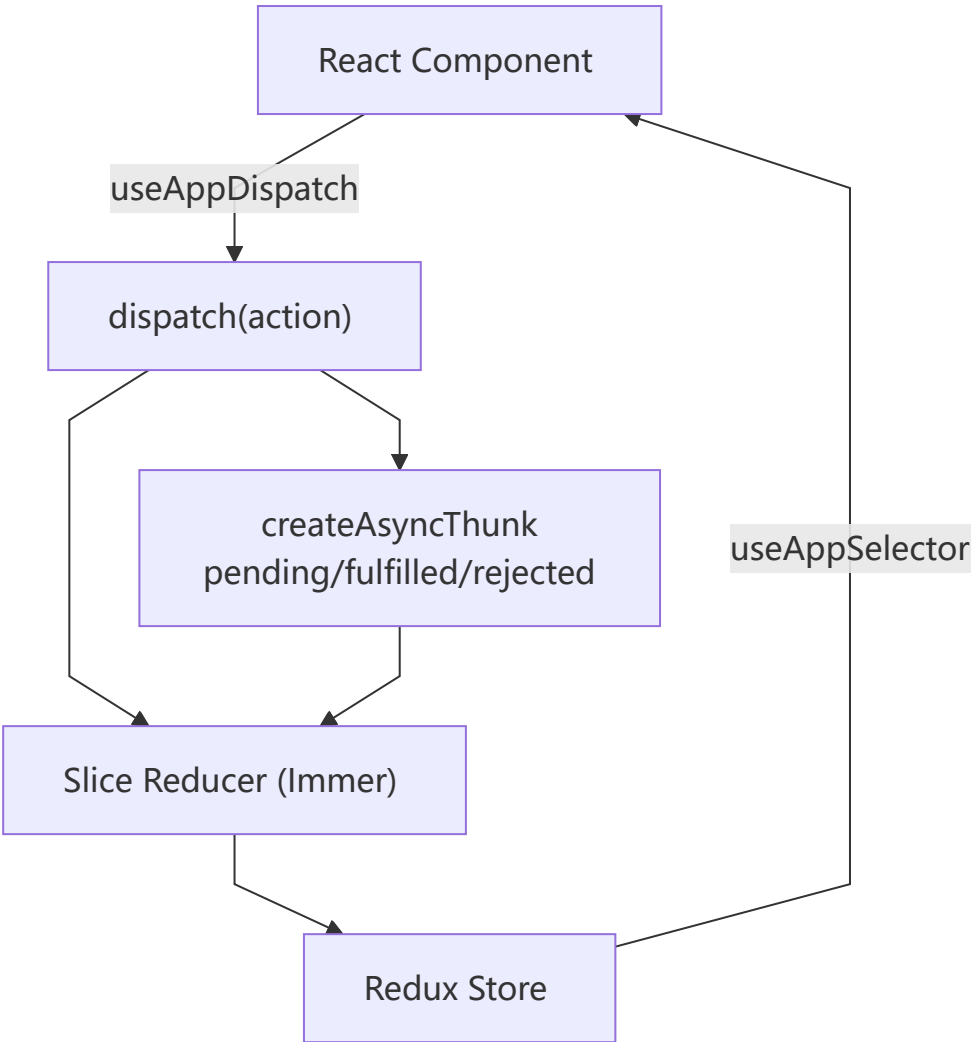
```
import { createAsyncThunk, createSlice } from '@reduxjs/toolkit';

interface ProductsState {
  items: Product[];
  loading: boolean;
  error: string | null;
}

export const fetchProducts = createAsyncThunk(
  'products/fetchProducts',
  async (_, { rejectWithValue }) => {
    try {
      const response = await fetch('/api/products');
      if (!response.ok) throw new Error('Failed to fetch');
      return await response.json() as Product[];
    } catch (err) {
      return rejectWithValue((err as Error).message);
    }
  }
);

const productsSlice = createSlice({
  name: 'products',
  initialState: { items: [], loading: false, error: null } as ProductsState,
  reducers: {},
  extraReducers: (builder) => {
    builder
      .addCase(fetchProducts.pending, (state) => {
        state.loading = true;
        state.error = null;
      })
      .addCase(fetchProducts.fulfilled, (state, action) => {
        state.loading = false;
        state.items = action.payload;
      })
      .addCase(fetchProducts.rejected, (state, action) => {
        state.loading = false;
        state.error = action.payload as string;
      });
  },
});
```

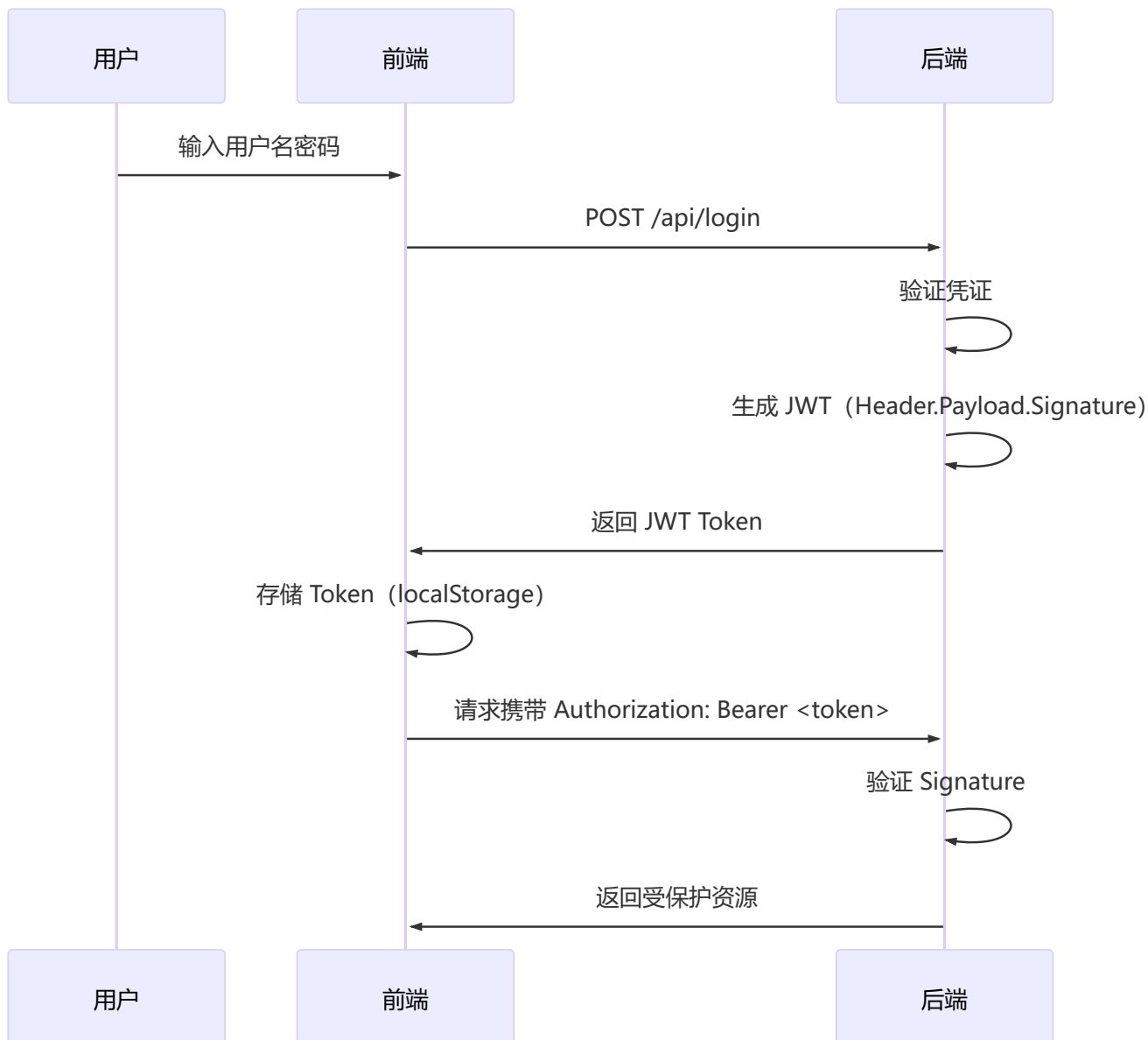
Redux Toolkit 数据流





第14章：全栈实战与电商系统

14-2 JWT 原理剖析



JWT 结构

```
Header:    { "alg": "HS256", "typ": "JWT" }
Payload:   { "sub": "user123", "iat": 1516239022, "exp": 1516242622 }
Signature: HMACSHA256(base64UrlEncode(header) + "." + base64UrlEncode(payload), secret)
```

14-4 注册与登录自定义 Hook

```
interface AuthState {
  user: User | null;
  token: string | null;
  loading: boolean;
  error: string | null;
```

```

}

function useAuth() {
  const [state, setState] = useState<AuthState>(() => {
    const token = localStorage.getItem('token');
    const user = localStorage.getItem('user');
    return {
      token,
      user: user ? JSON.parse(user) : null,
      loading: false,
      error: null,
    };
  });

  const login = useCallback(async (email: string, password: string) => {
    setState(prev => ({ ...prev, loading: true, error: null }));
    try {
      const res = await fetch('/api/auth/login', {
        method: 'POST',
        headers: { 'Content-Type': 'application/json' },
        body: JSON.stringify({ email, password }),
      });
      if (!res.ok) throw new Error('登录失败');
      const data = await res.json();
      localStorage.setItem('token', data.token);
      localStorage.setItem('user', JSON.stringify(data.user));
      setState({ user: data.user, token: data.token, loading: false, error: null });
      return true;
    } catch (err) {
      setState(prev => ({ ...prev, loading: false, error: (err as Error).message }));
      return false;
    }
  }, []);

  const logout = useCallback(() => {
    localStorage.removeItem('token');
    localStorage.removeItem('user');
    setState({ user: null, token: null, loading: false, error: null });
  }, []);

  return { ...state, login, logout, isAuthenticated: !!state.token };
}

```

14-5 路由守卫 + Token 管理

```

function AuthGuard({ children }: { children: React.ReactNode }) {
  const { isAuthenticated, token } = useAuth();

  if (!isAuthenticated) {
    return <Navigate to="/login" state={{ from: location }} replace />;
  }
}

```

```
// Token 过期检查
if (token) {
  const payload = JSON.parse(atob(token.split('.')[1]));
  if (payload.exp * 1000 < Date.now()) {
    return <Navigate to="/login" replace />;
  }
}

return <>{children}</>;
}

// 路由配置
<Route path="/checkout" element={
  <AuthGuard>
    <CheckoutPage />
  </AuthGuard>
} />
```

14-8 本地存储 vs 后端数据：联网版购物车

```
function useSyncCart() {
  const { isAuthenticated } = useAuth();
  const localCart = useLocalStorage<CartItem[]>('cart', []);
  const [serverCart, setServerCart] = useState<CartItem[]>([]);

  // 登录后：将本地购物车同步到服务器
  useEffect(() => {
    if (isAuthenticated && localCart[0].length > 0) {
      syncCartToServer(localCart[0]);
      localCart[1]([]);
      fetchServerCart().then(setServerCart);
    }
  }, [isAuthenticated]);

  const items = isAuthenticated ? serverCart : localCart[0];
  const setItems = isAuthenticated ? setServerCart : localCart[1];

  return { items, setItems };
}
```

14-10 从购物车到支付成功

```
function CheckoutFlow() {
  const { items, totalAmount } = useCart();
  const [step, setStep] = useState<'cart' | 'shipping' | 'payment' | 'confirm'>('cart');
  const [orderId, setOrderId] = useState<string | null>(null);

  const handlePlaceOrder = async () => {
    try {
      const res = await fetch('/api/orders', {
        method: 'POST',

```

```

    headers: {
      'Content-Type': 'application/json',
      Authorization: `Bearer ${localStorage.getItem('token')}`,
    },
    body: JSON.stringify({ items, totalAmount }),
  });
  const order = await res.json();
  setOrderId(order.id);
  setStep('confirm');
} catch (err) {
  console.error('下单失败', err);
}
};

return (
  <div className="max-w-2xl mx-auto p-6">
    {step === 'cart' && <CartStep onNext={() => setStep('shipping')} />}
    {step === 'shipping' && <ShippingForm onNext={() => setStep('payment')} />}
    {step === 'payment' && <PaymentForm onSubmit={handlePlaceOrder} />}
    {step === 'confirm' && <OrderConfirmation orderId={orderId} />}
  </div>
);
}

```



高频面试题精选

Q1: React 渲染流程 (Trigger→Render→Commit)

React 的渲染分为三个阶段：

- **Trigger**: setState / useState 触发更新
- **Render**: 构建虚拟 DOM 树，执行 Diff
- **Commit**: 将差异应用到真实 DOM



Q2: useEffect 执行时序

阶段	执行内容
挂载	render → DOM 更新 → 浏览器绘制 → useEffect 执行
更新	render → DOM 更新 → 浏览器绘制 → 清理旧 effect → 执行新 effect
卸载	清理 effect

Q3: React 19 Actions 机制

Actions 是 React 19 的统一表单处理方案，自动管理 pending 状态、错误处理和乐观更新。

```
async function submitAction(prevState: any, formData: FormData) {
  'use server';
  // 处理表单提交
}

function Form() {
  const [state, formAction] = useActionState(submitAction, null);
  const { pending } = useFormStatus();
  return <form action={formAction}>...</form>;
}
```

Q4: key 的作用

key 帮助 React 识别列表中的每个节点，用于 Diff 算法中的节点复用判断。不设 key 或使用 index 作为 key 会导致：

- 列表排序时组件状态错乱
- 不必要的 DOM 操作
- 潜在的性能问题

Q5: Concurrent Mode 并发模式

React 18+ 的并发模式允许渲染过程可中断，高优先级任务（用户输入）可以打断低优先级任务（数据加载）。

```
import { startTransition } from 'react';

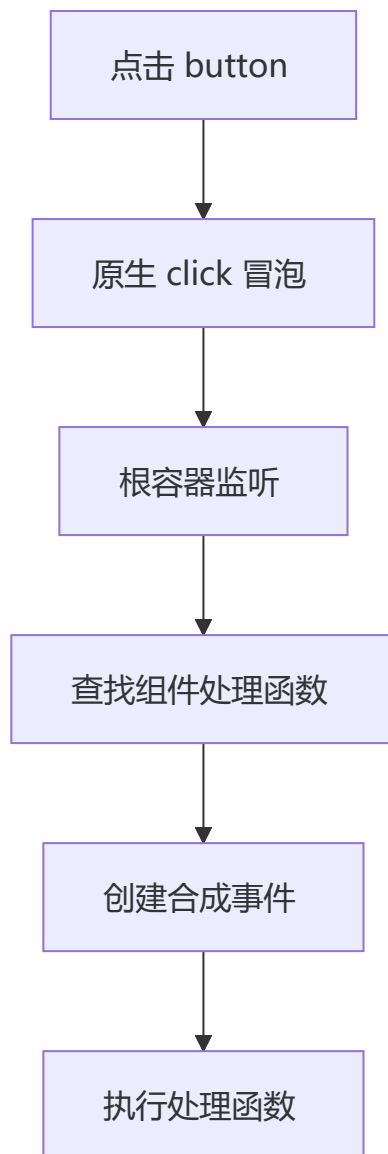
startTransition(() => {
  setSearchResults(filterData(input));
});
```

Q6: use() vs useEffect

特性	use()	useEffect
引入版本	React 19	React 16.8
调用位置	组件顶层/条件中	组件顶层
数据来源	Promise / Context	任意副作用
配合 Suspense	✅ 必须	❌
条件调用	✅ 允许	❌ 禁止

Q7: React 合成事件

React 的事件并非绑定在真实 DOM 上，而是通过事件代理统一绑定在根容器上。



Q8: Hooks 条件调用限制

原因：React 使用**链表**存储 Hooks 状态，每次渲染必须按**相同顺序**调用，否则链表顺序被打乱导致状态错乱。

```
// 链表结构
fiber.memoizedState → Hook1(state) → Hook2(state) → Hook3(state) → null
```

Q9: Server Component vs Client Component

特性	Server Component	Client Component
渲染位置	服务器	客户端浏览器
交互能力	✗ 无交互	✓ 有交互
数据访问	✓ 直接访问 DB	✗ 通过 API
Bundle	✗ 不包含	✓ 包含

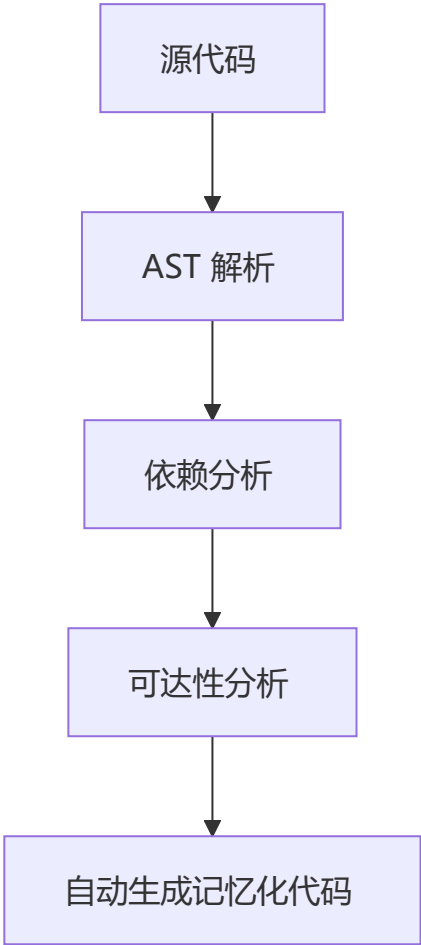
特性	Server Component	Client Component
指令	默认	'use client'

Q10: React.memo vs useMemo

维度	React.memo	useMemo
对象	组件	值/计算结果
作用	跳过组件重渲染	跳过昂贵计算
对比方式	浅比较 props	依赖数组
返回值	组件	计算值

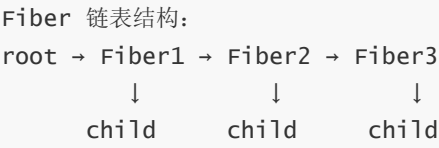
Q11: React Compiler 原理

React Compiler（原 Forget）在**编译阶段**进行静态分析，自动推导值的依赖关系，生成记忆化代码，消除手动 useMemo/useCallback 的需求。



Q12: Fiber 可中断渲染

Fiber 架构将渲染任务拆分为小单元，每执行一个单元后检查是否有更高优先级任务。



Q13: React 事件机制对比

对比项	原生事件	React 事件
绑定方式	<code>addEventListener</code>	<code>onClick={handler}</code>
命名规则	全小写	小驼峰
默认行为	<code>return false</code>	<code>e.preventDefault()</code>
事件对象	原生 Event	合成事件 SyntheticEvent
代理节点	目标元素	根容器 (React 17+)